

Artem Semidetnov

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Profile

Mathematician and research engineer working across category theory, dependent type theory, proof assistants, and AI-assisted mathematical research. I build formalizations and programming-language tooling, from typecheckers and IDE integrations to LLM agents for formal mathematics.

Technical Skills

- **Formal methods and mathematics:** category theory, homotopy type theory, dependent type theory, proof assistants, theorem proving, formalization.
- **Programming and language tooling:** Haskell, Kotlin, Java, Python; typechecking, compiler-style tooling, IntelliJ Platform.
- **AI engineering:** MCP, LLM coding agents, model fine-tuning, code completion, evaluation.
- **Languages:** Russian (native), English C1 (IELTS 8.0/9), French (conversational).

Selected Research and Engineering Projects

Rzk proof assistant and sHoTT library [↗](#)

Open source
Aug 2026 – present

- Contribute to an experimental proof assistant for synthetic infinity-categories across its Haskell typechecker and Rzk formalization library.
- Implemented identity-type inference from either endpoint, expanded error diagnostics, and added regression tests in an [open pull request](#) [↗](#).
- Formalized that precomposition with an equivalence is an equivalence in the [sHoTT library](#) [↗](#).

Work Experience

JetBrains, Research Intern in HoTT and Dependent Types Lab

Remote
Nov 2025 – Apr 2026

- Developed an MCP server for the Arend proof assistant, exposing typechecking and IDE functionality to LLM coding agents.
- Integrated the server with the Arend plugin for IntelliJ IDEA and contributed plugin changes.
- Built an experimental LLM coding agent for Arend using Koog; fine-tuned models for Arend code completion.

JetBrains, Research Intern in HoTT and Dependent Types Lab

Pafos, Cyprus
June 2024 – Sept 2024

- Contributed formalized results in algebra and homotopy type theory to the official Arend library.
- Formalized the Eckmann-Hilton argument, Eilenberg-MacLane spaces, homology of types, group automorphisms, Schur's Lemma, Maschke's Lemma, and characterizations of group actions; selected work shipped in [Arend 1.10](#) [↗](#).

IPONWEB (acquired by Criteo), Machine Learning Intern

Pafos, Cyprus
May 2023 – Sept 2023

- Reverse-engineered a Criteo ML system that analyzes websites and produces word2vec representations; evaluated potential applications.

Education

MSc **University of Geneva**, Mathematics

2025 – 2027

BSc **Saint Petersburg State University**, Mathematics

2021 – 2025

Research Interests

- Categorical semantics of computation; categorical homotopy theory; simplicial methods and operads; homotopy type theory and computer-verified mathematics; AI-assisted mathematical reasoning; geometric and combinatorial group theory.

Publications

The operad associated to a crossed simplicial group [↗](#)

Dec 2025

Artem Semidetnov · [doi:10.48550/arXiv.2512.14498](https://doi.org/10.48550/arXiv.2512.14498) [↗](#)

On the geometry of free nilpotent groups [↗](#)

May 2021

Artem Semidetnov, Ruslan Magdiev · [doi:10.48550/arXiv.2106.00095](https://doi.org/10.48550/arXiv.2106.00095) [↗](#)

AI-Assisted Mathematical Research

- [An explicit non-elementary matrix over a bivariate Laurent polynomial ring](#) [↗](#) - Vasiliy Ionin, **Artem Semidetnov** (Sept 2026).
- [On Some More Problems from the Kourovka Notebook](#) [↗](#) - Vasiliy Ionin, **Artem Semidetnov** (Aug 2026).
- [The stable commutator length of a relator is not a one-relator group invariant](#) [↗](#) - **Artem Semidetnov** (Aug 2026).

Awards and Selected Activities

[Excellence Master Fellowship at the University of Geneva](#) [↗](#)

JetBrains Scholarship at Neapolis University Pafos

Funded participant at [Teaching Higher Category Theory with Computers](#) [↗](#), ICERM, Brown University, 2026.

Selected Talks

The operad associated to a crossed simplicial group [↗](#)

July 2026

- Young Topologists Meeting 2026, Copenhagen

Plus construction and simplicial methods [↗](#)

Aug 2025

- Algebraic Topology, Geometry, Combinatorics, and Data Analysis, Pushkin

Twisting numbers on braid and Thompson groups [↗](#)

2024

- Topological Methods in Dynamics and Related Topics VII, Nizhny Novgorod